QP Code **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION 2026

FIFTH SEMESTER

ELECTIVE- MG5DSEBTS301- INTRODUCTION TO FORENSIC BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The principle that "every contact leaves a trace" was proposed by: [K] / [CO1]
   a) Hans Gross b) Edmond Locard c) Francis Galton d) William Herschel

2. _____ in forensic science mainly deals with the examination of Blood and other biological fluids [K] / [CO3]
   a) Serology b) Toxicology c) Anthropology d) Chemistry

3. Mass spectrometry identifies compounds primarily based on their: [K] / [CO5]
   a) Colour b) Mass-to-charge ratio c) Solubility d) Melting point

4. Which technique is commonly used to identify an individual based on their DNA ? [K / [CO2]
   a) Chromatography b) DNAprofiling c) Spectroscopy d) Centrifugation

5. Which technique is commonly used for the identification of unknown chemical compounds? [K] / [CO3]
   a) Chromatography b) Microscopy c) Electrophoresis d) Centrifugation

6. Forensic toxicology primarily deals with the detection and identification of: [K] / [CO2]
   a) Drugs and toxic substances b) Microorganisms c) Plant pigments d) Proteins

7. Toxicological findings are primarily interpreted by considering the concentration of a substance along with: [U] / [CO5]
   a) Clinical and circumstantial evidence b) Sample colour alone c) Body weight alone d) Age of the laboratory instrument

8. Which organ is primarily responsible for the metabolism of many drugs? [K] / [CO3]
   a) Heart b) Liver c) Lung d) Spleen

9. The scientific study of human skeletal remains in a forensic context is known as: [K] / [CO1]
   a) Forensic anthropology b) Forensic toxicology c) Forensic chemistry d) Forensic serology

10. Maintaining the documented history of the handling of evidence is known as: [K] / [CO4]
    a) Chain of custody b) Cross-examination c) Peer review d) Profiling

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. What factors should be considered while interpreting biological evidence in a criminal investigation? [U] / [CO3]

12. Briefly describe the historical development of forensic science. [U / [CO1]

13. What is spectroscopy? Mention any two applications of spectroscopy in forensic analysis. [K / [CO5]

14. What is forensic biology? Mention any two types of biological evidence examined in forensic investigations. [K] / [CO2]

15. Define forensic toxicology and mention any two biological samples used for toxicological analysis. [K] / [CO2]

16. Mention any three techniques used for the identification of drugs in forensic samples. [U] / [CO5]

17. Explain the effects of drug use on the central nervous system. [U] / [CO3]

18. Define forensic anthropology and mention any two applications in criminal investigations. [K] / [CO1]

19. Mention any three skeletal features used for estimating sex from human skeletal remains. [K] / [CO5]

20. What is chain of custody? Mention its importance in forensic investigations. [K] / [CO4]

21. Mention any three analytical techniques used in forensic chemistry and state their applications. [K] / [CO3]

22. Explain the role of a forensic scientist as an expert witness in court. [U] / [CO4]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Describe the basic steps involved in DNA profiling and explain its importance in criminal investigations. [U / [CO2]

24. Explain how chromatographic techniques are used for the identification and analysis of drugs in forensic investigations. [U] / [CO5]

25. Explain the principles and forensic applications of serology in the identification and analysis of biological evidence. [U] / [CO3]

26. Explain the significance of human skeletal remains in the identification of unknown individuals during criminal investigations. [U] / [CO1]

27. Describe the major analytical techniques used for the identification of drugs in forensic samples. [U] / [CO5]

28. Describe how drugs affect the central nervous and hepatic systems. [U] / [CO3]

29. Discuss the ethical responsibilities of forensic scientists. [U] / [CO4]

30. Explain the major legal considerations involved in the collection, preservation, analysis, and presentation of forensic evidence. [U] / [CO4]

[5x6 = 30]
